



Batterjee Medical College
For Science and Technology

Lung Cancer

An approach to a Case of Hemoptysis

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Main Subtopics



- Epidemiology
- Etiology
- Types
- Staging
- Clinical Manifestations
- Diagnostics
- Management
- Treatment
- Solitary Lung Nodules

Learning Objectives

Knowledge:

- Describe the epidemiology and risk factors of lung cancer, including the impact of smoking, radon, asbestos, and other environmental carcinogens.
- Classify the major histological subtypes of lung cancer and identify their primary anatomical locations and cell origins.
- Detail the clinical presentation of lung cancer, including pulmonary, constitutional, local invasive, and metastatic manifestations.
- Outline the classical paraneoplastic syndromes associated with specific lung cancer subtypes.
- Differentiate between the TNM staging system used for NSCLC and the Limited vs. Extensive stage system used for SCLC.
- Formulate the differential diagnosis and initial diagnostic workup for a patient presenting with hemoptysis.
- Identify current screening guidelines for lung cancer using low-dose CT.

Skills:

- Interpret diagnostic modalities including chest X-rays and chest CT scans to identify pulmonary masses and solitary lung nodules.
- Formulate a systematic diagnostic algorithm for evaluating a solitary pulmonary nodule.
- Construct an initial diagnostic approach to hemoptysis, distinguishing between non-massive and massive cases.
- Select appropriate invasive and non-invasive diagnostic procedures to achieve definitive tissue diagnosis and staging.
- Apply evidence-based management principles to outline appropriate treatment modalities based on cancer stage and histological subtype.
- Execute life-saving resuscitation and airway management strategies in cases of massive hemoptysis.

Values:

- Demonstrate a professional attitude in adherence to institutional and hospital training rules and regulations during clinical exposure.
- Value the role of multidisciplinary care, evidence-based guidelines, and early screening interventions in improving patient outcomes and reducing lung cancer mortality.

Lung Cancer

Risk Factors:

- **Smoking**

Associated with the development of 80–90% of lung cancers

Can be related to the number of pack - years

- **Occupational and environmental exposure to carcinogens**

Passive smoking

Radon (2nd leading cause of lung cancer) and uranium

Asbestos

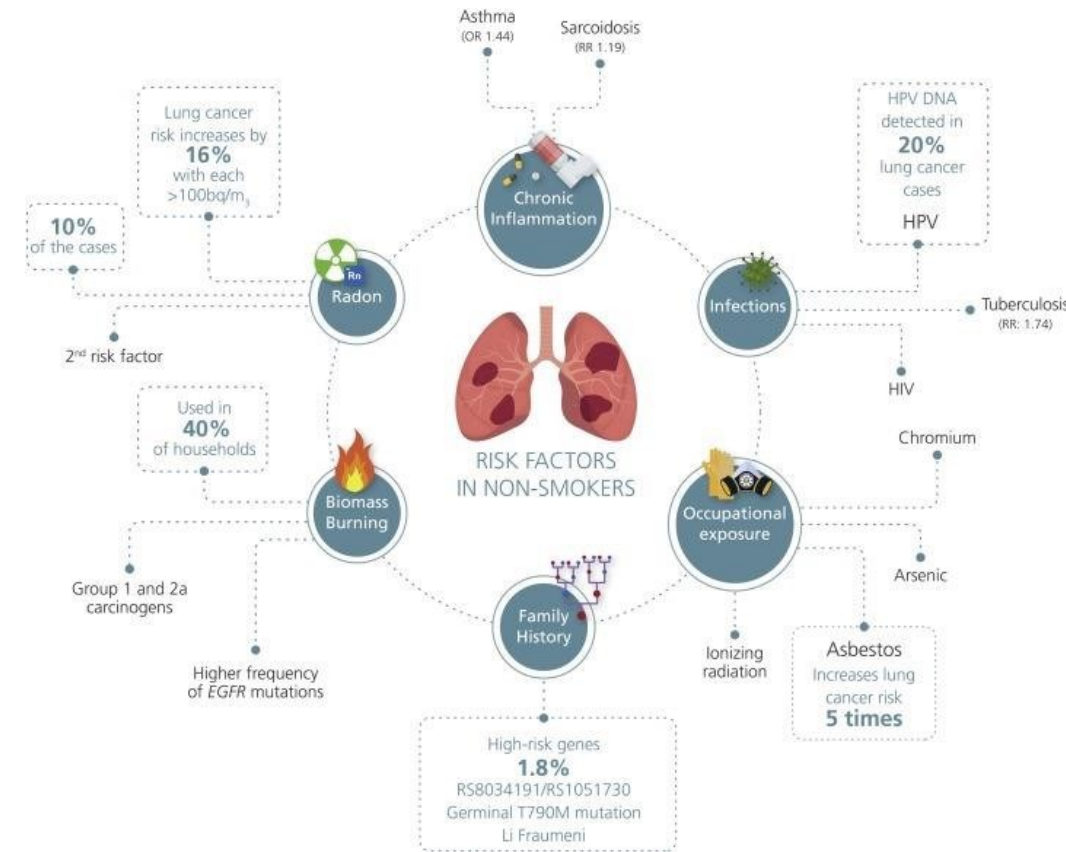
Occupational carcinogens (e.g., arsenic, silica)

Environmental air pollution

- **Family history** (genetic predisposition)

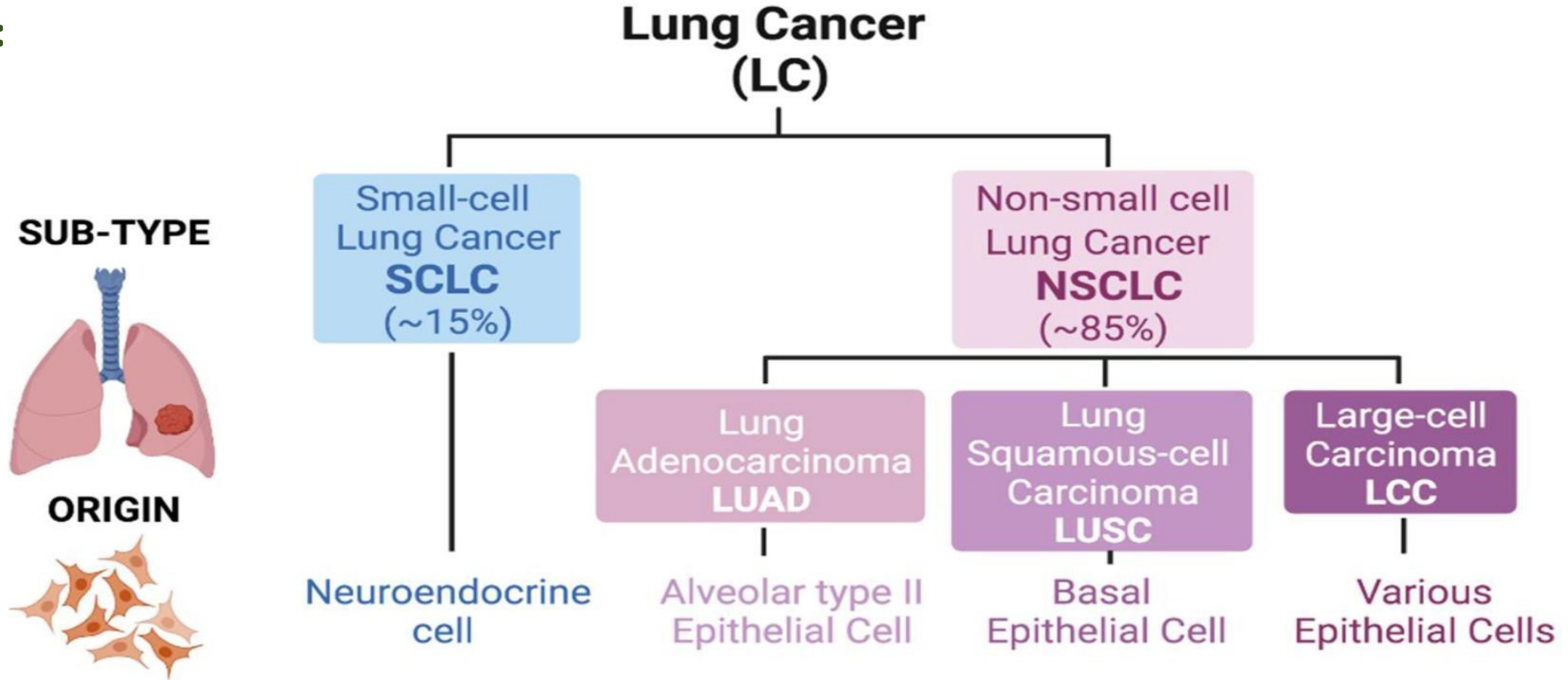
- **Other risk factors:**

pulmonary scarring, previous radiation, pulmonary fibrosis, chronic infections (e.g., tuberculosis, HIV)



Lung Cancer

Types:



Lung Cancer

Tumor type	Location	Characteristics
Non-small cell lung cancer		
Adenocarcinoma	•Peripheral	•Most common type of primary lung cancer •More common in women and nonsmokers •Common finding: hypertrophic osteoarthropathy •Most common type of lung cancer that originates in pulmonary scars •Prognosis is usually better than in other types
Squamous cell carcinoma	•Central	•Strong association with smoking •Cavitary lesions arising from a hilar bronchus •PTHrP: hypercalcemia (see “Paraneoplastic syndromes”)
Large cell carcinoma	•Peripheral	•Strong association with smoking •Poor response to chemotherapy •Early metastases •Poor prognosis
Lung neuroendocrine tumors		
Small cell lung cancer	•Central	•Strong association with smoking •Associated with several paraneoplastic syndromes •Undifferentiated and very aggressive with early metastases

Lung Cancer

Staging:

NSCLC is staged via the primary TNM system.

T1 	• Tumor ≤ 3 cm • Bronchoscopically visible invasion distal to lobar bronchus • Superficial spreading tumor of any size, invasion limited to bronchial wall	 N1: Ipsilateral hilar, intrapulmonary, and/or peribronchial N2: Subcarinal, ipsilateral mediastinal N3: Contralateral mediastinal and hilar, ipsilateral or contralateral supraclavicular or scalene
T2 	• Tumor >3 cm to ≤ 5 cm • Involves main bronchus, without main carinal involvement, with atelectasis and/or obstructive pneumonia of part or all of lung • Invades visceral pleura • Extends across fissure or involves two adjacent lobes	
T3 	• Tumor >5 cm to ≤ 7 cm • Parietal pericardium or phrenic nerve invasion • Invades parietal pleura • Separate tumor nodules in the same lobe as primary tumor • Chest wall invasion	
T4 	• Tumor >7 cm • Invades trachea, recurrent laryngeal nerve, great vessels, diaphragm, esophagus, and/or vertebral body • Involves main carina • Separate tumor nodules in a different ipsilateral lobe 	

Summary of Lung Cancer Staging IASLC 8th Edition

Summary of the 8th Edition of the TNM Classification for Thoracic Cancers. IASLC. 2018. www.iaslc.org/research-education/publications-resources-guidelines/summary-8th-edition-tnm-classification

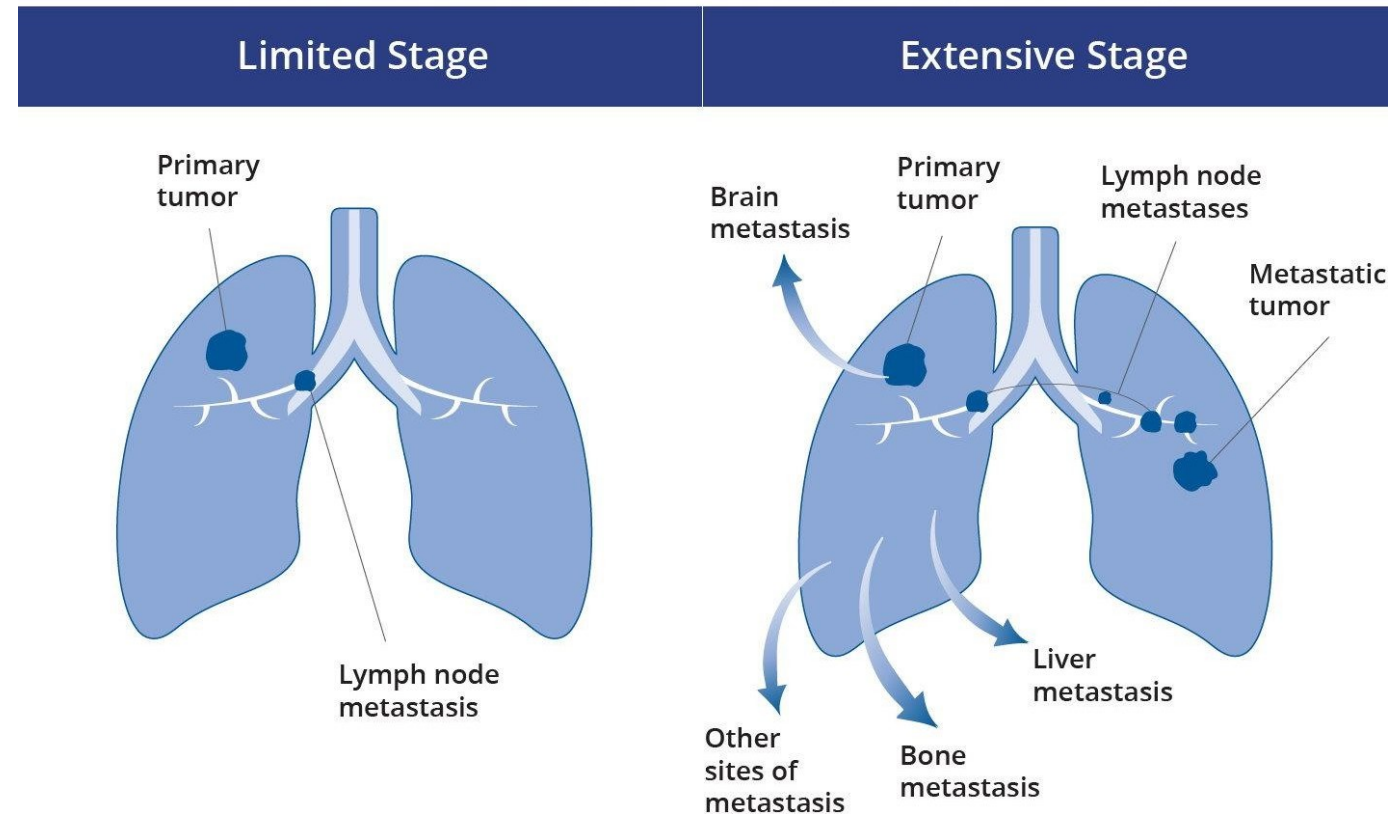


Lung Cancer

Staging:

- SCLC is staged differently
- Limited—confined to chest plus supraclavicular nodes, but not cervical or axillary nodes
- Extensive—outside of chest and supraclavicular nodes

Small Cell Lung Cancer (SCLC) Staging



Lung Cancer

Signs & Symptoms:

Can be divided into five main groups:

- Local invasion (SVC, Horner, Pancoast, etc.)
- Local manifestations (cough, hemoptysis, dyspnea, etc.)
- Constitutional symptoms (fatigue, weight loss, etc)
- Metastatic disease
- Paraneoplastic syndrome



COUGH



SHORTNESS of
BREATH



PLEURITIC
CHEST PAIN



UNINTENTIONAL
WEIGHT LOSS



FATIGUE



HEMOPTYSIS

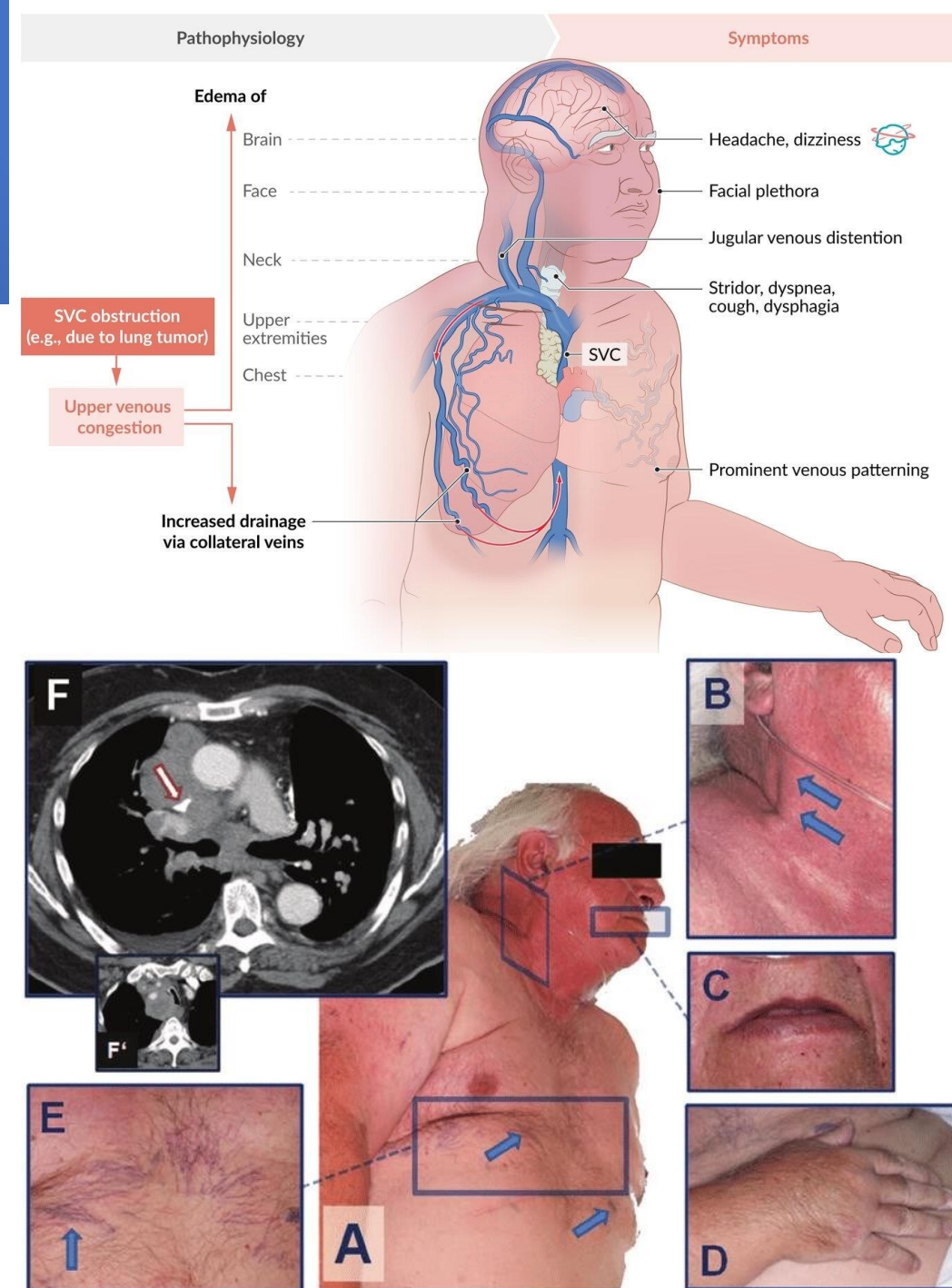


HISTORY of RECURRENT
EPISODES of PNEUMONIA

Variants Lung Cancer

Superior Vena Cava Syndrome:

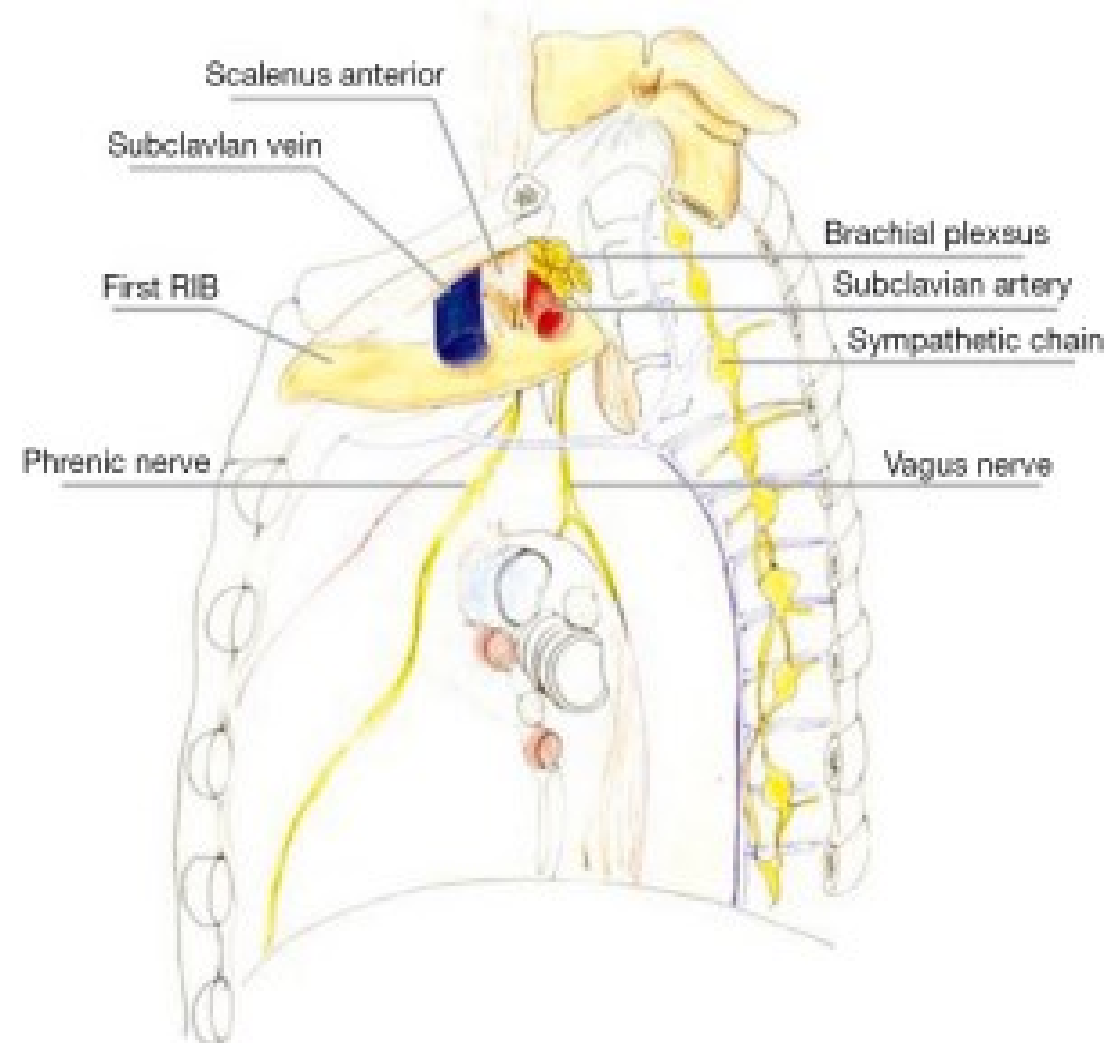
- Caused by obstruction of SVC by a mediastinal tumor
- More commonly occurs with SCLC
- drainage increases into the inferior vena cava via collateral veins, resulting in a prominent venous patterning on the upper body. Venous congestion proximally to the obstruction.
- Findings: facial fullness; facial and arm edema; dilated veins over anterior chest, arms, and face; jugular venous distention (JVD)
- Hoarseness of voice
- Dysphagia
- Post obstructive pneumonia
- Dyspnea (diaphragmatic paralysis)



Variants of Lung Cancer

Pancoast Tumor:

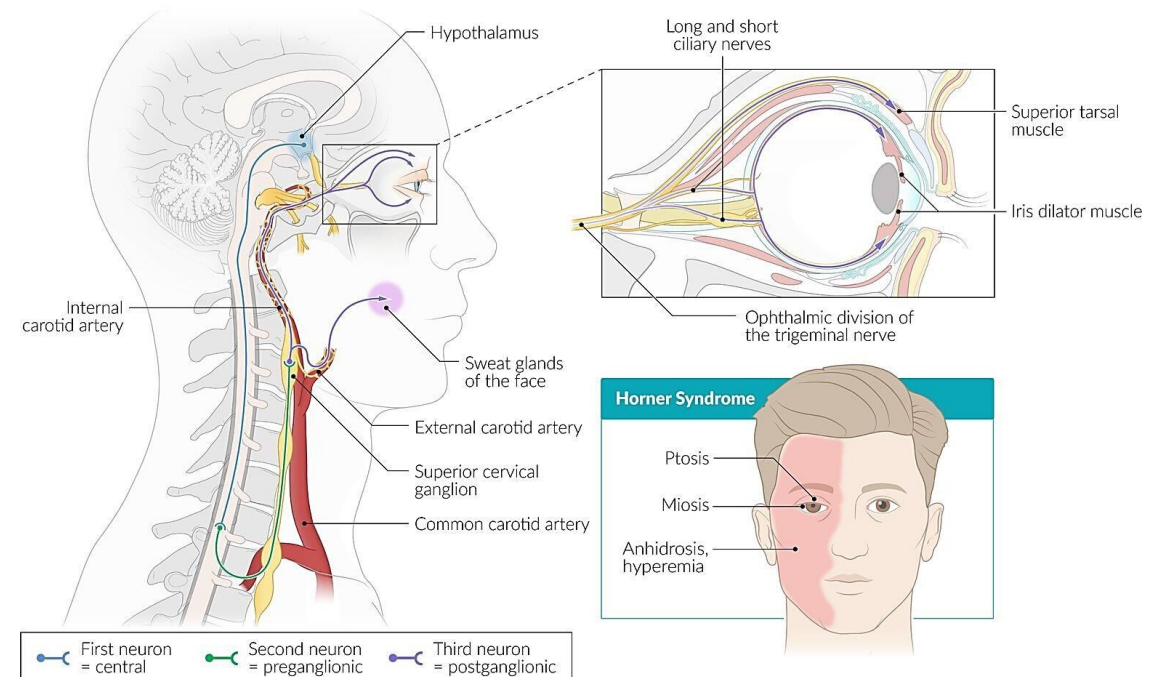
- Apical tumor located in the Superior sulcus involving C8 and T1–T2 nerve roots causing shoulder pain radiating down the arm
- Usually Non small lung cancer
- Leads to **Pancoast syndrome**, which is secondary to mass effect:
- **Pain, upper limb motor and sensory deficit** due to brachial plexus affection (may cause wasting of small muscles of the hand)
- **Horner syndrome**, due to affection of cervical sympathetic ganglion
- **Hoarseness of voice**: recurrent laryngeal nerve
- **Fascial swelling and unilateral edema** of arm due to brachiocephalic vein affection
- **Paralysis of hemidiaphragm** due to phrenic nerve affection



Lung Cancer

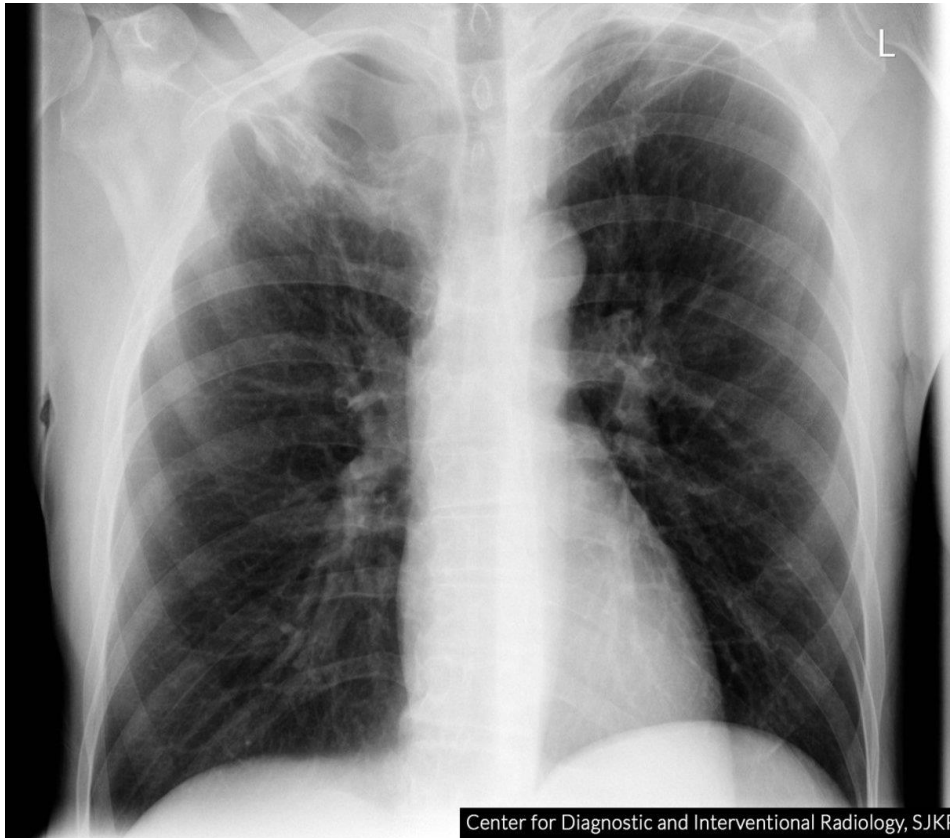
Horner Syndrome:

- Due to invasion of cervical sympathetic chain by an apical tumor
- Symptoms: unilateral facial anhidrosis (no sweating), ptosis, and miosis



Lung Cancer

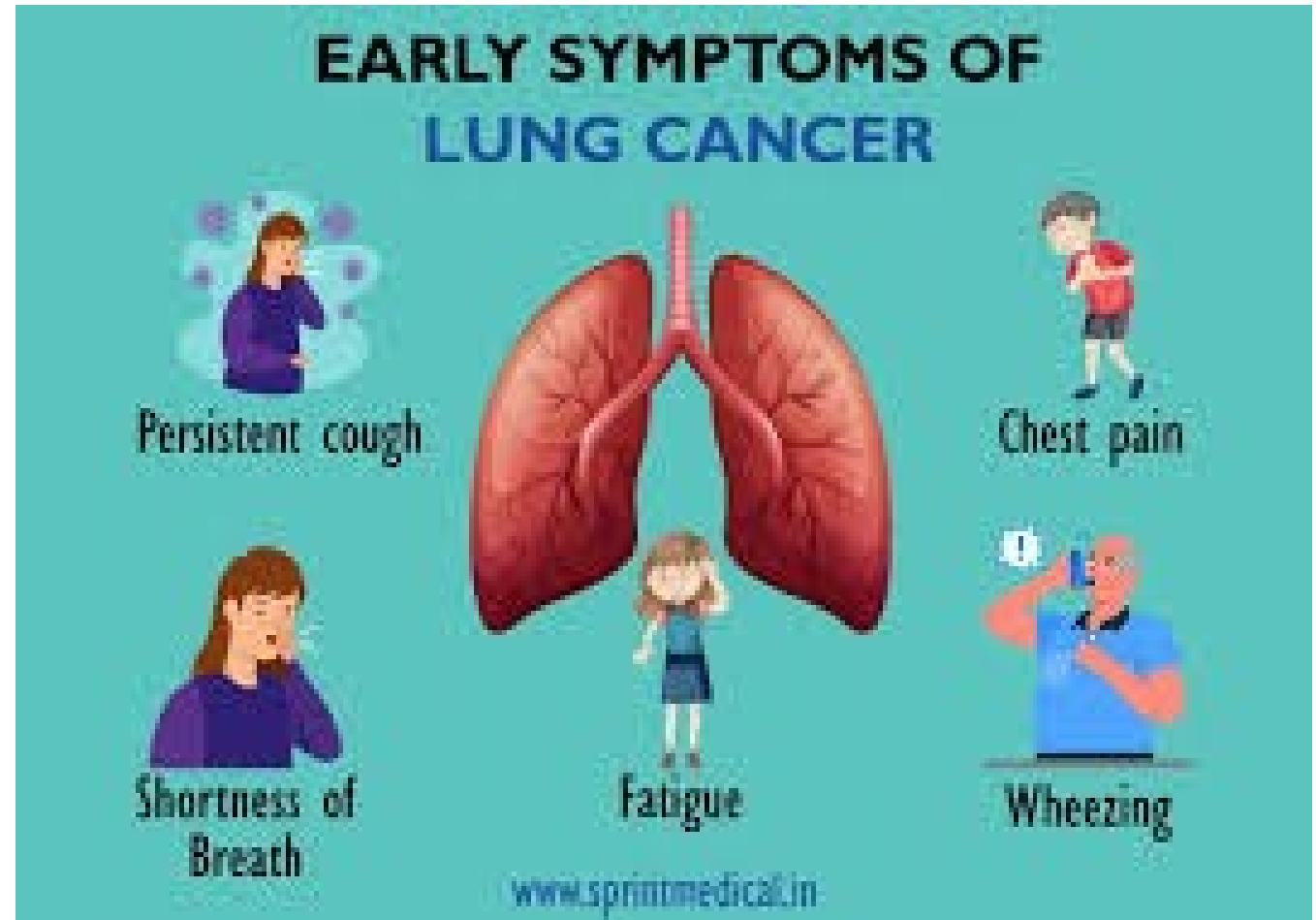
Pancoast syndrome:



Clinical picture

Pulmonary symptoms

- Cough, hemoptysis
- Progressive dyspnea
- Wheezing
- Chest pain



Clinical picture

Extrapulmonary symptoms

- **Constitutional symptoms:**

weight loss, fever, weakness

Signs and symptoms of tumor infiltration and/or compression of neighboring structures

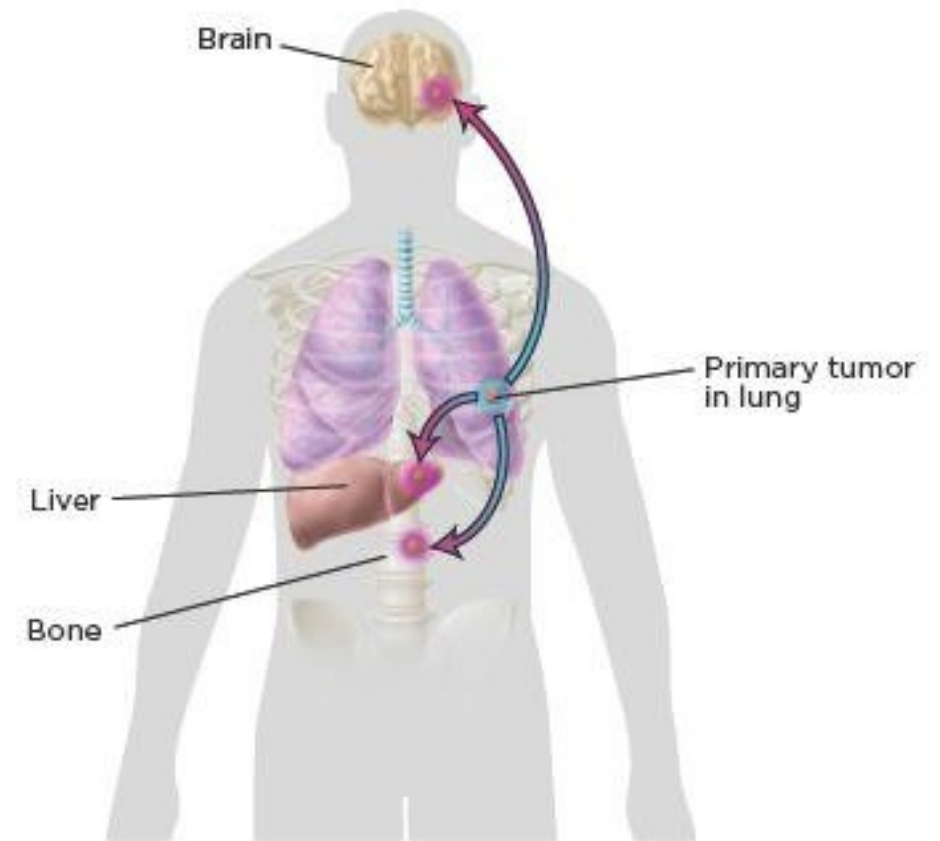
- Superior vena cava syndrome
- Hoarseness of voice
- Dyspnea and diaphragmatic elevation
- Dullness on percussion, reduced breath sounds: malignant pleural effusion on the affected side
- Post obstructive pneumonia
- Dysphagia: esophageal compression

Lung Cancer

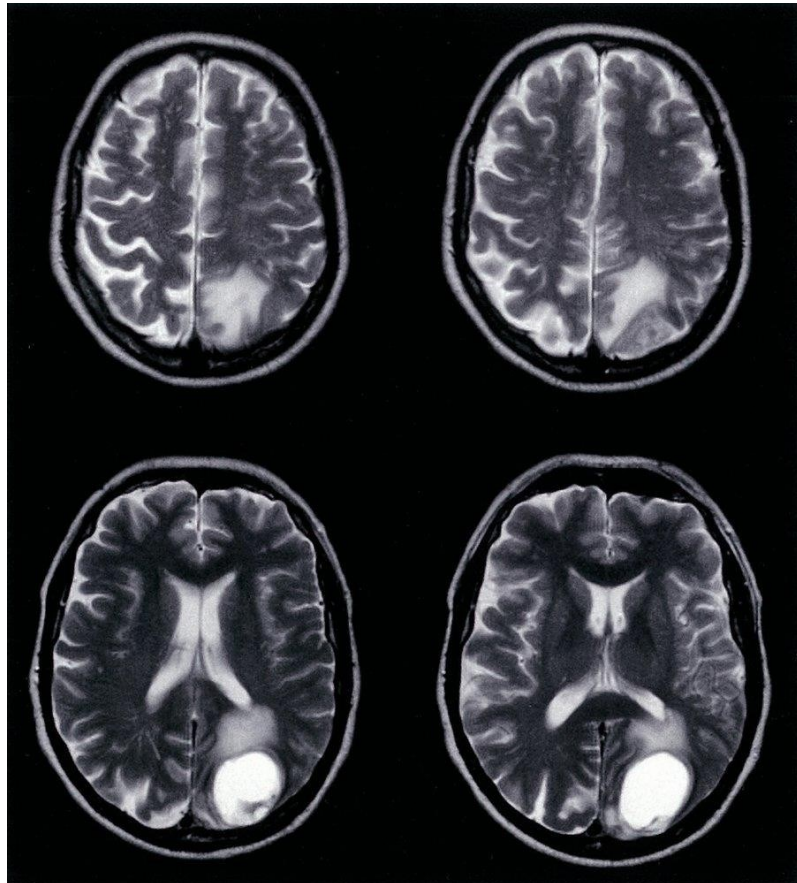
Symptoms of metastatic disease:

- Brain: headaches, seizures, focal motor deficits
- Liver: nausea, jaundice, ascites
- Adrenal gland: typically asymptomatic
- Bones: bone pain, elevated serum alkaline phosphatase and calcium

Lung Cancer Metastasis



Lung Cancer



Lung Cancer

Paraneoplastic Syndromes & Complications

SMALL CELL LUNG CANCER

Ectopic Cushing Syndrome



ACTH -
MOON FACE

SIADH (endocrine-mediated)

Lambert-Eaton myasthenic syndrome &
Other immune-mediated
neurological syndromes



LOW URINE
OUTPUT

ADENOCARCINOMA

Nonbact. verrucous endocarditis



SQUAMOUS CELL CARCINOMA

Hypercalcemia (PTHrP)

Associated with poor prognosis.



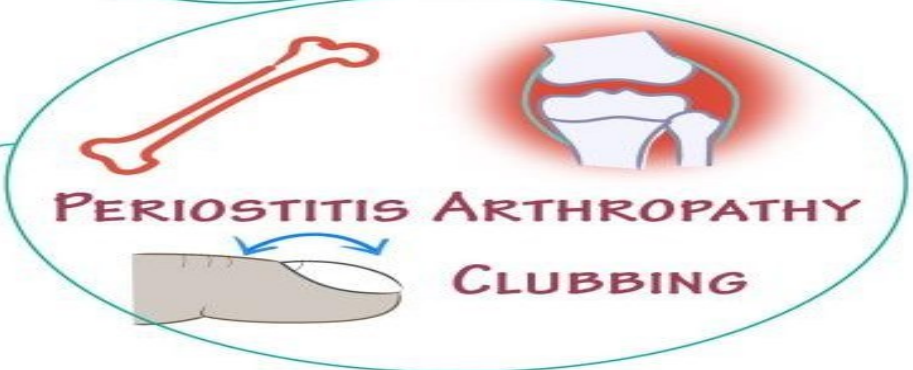
LARGE CELL LUNG CANCER

Gynecomastia



NSCLC

Hypertrophic pulmonary osteoarthropathy
(aka, Marie -Bamberger syndrome) Rare.



PERIOSTITIS ARTHROPATHY

CLUBBING

ALL LUNG CANCERS

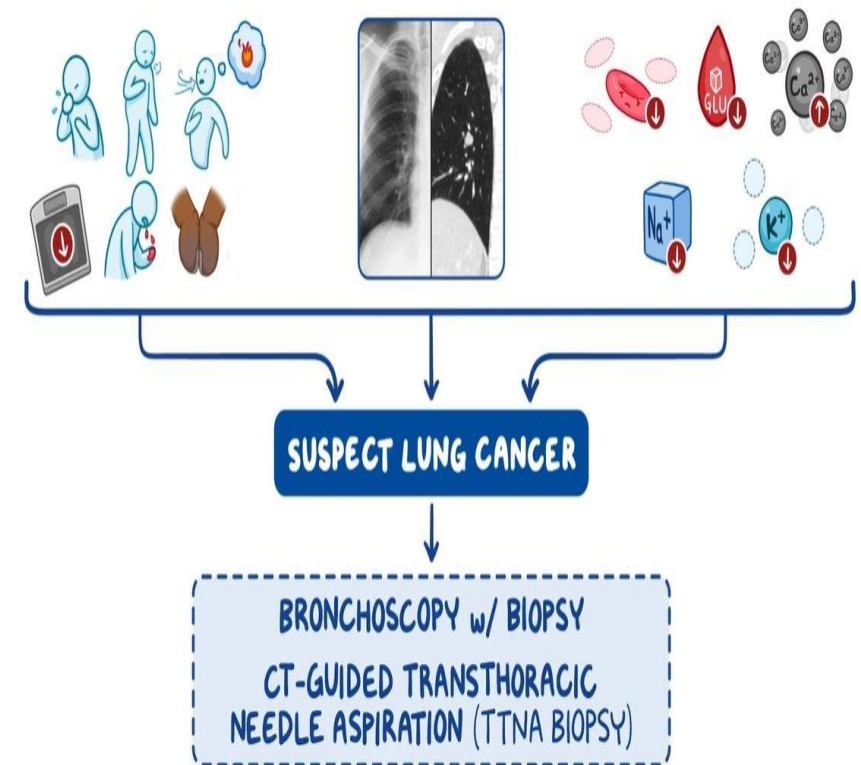
Anemia, DIC,
Granulocytosis, thrombocytosis.
Dermatomyositis



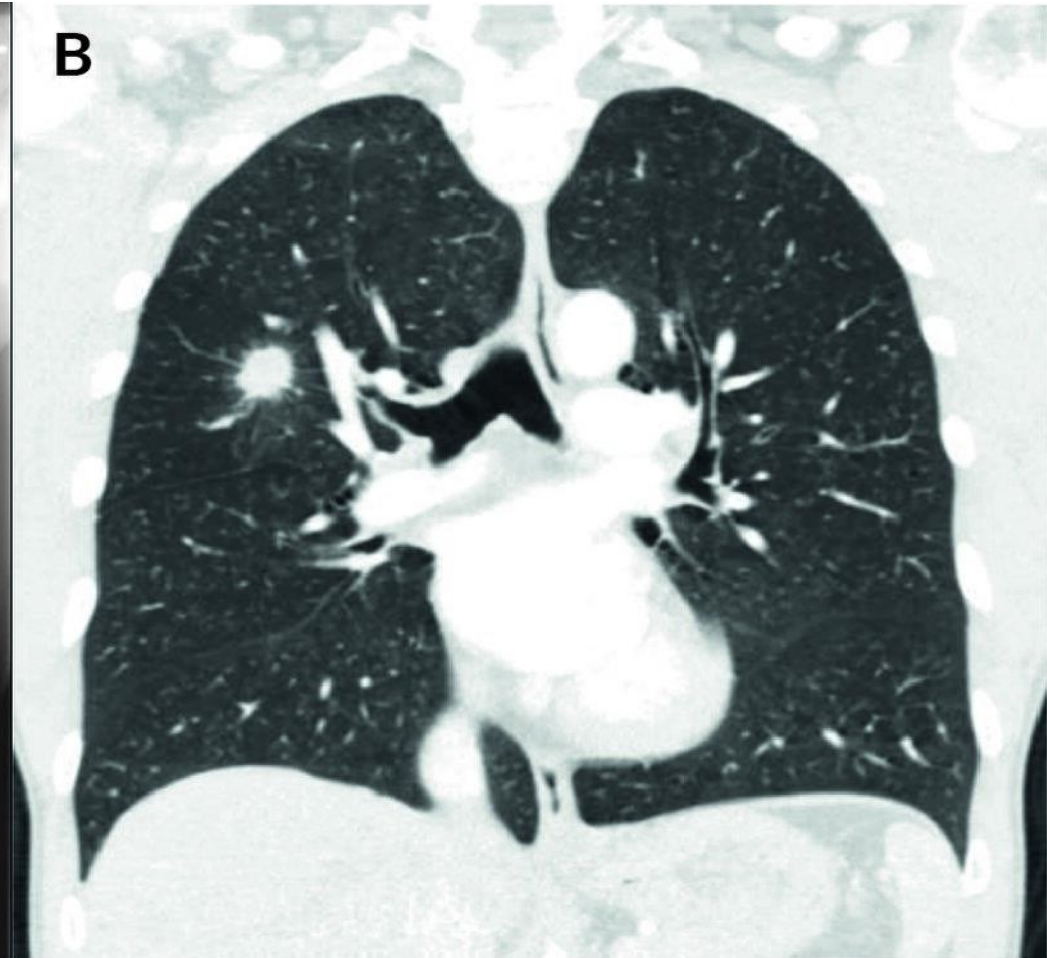
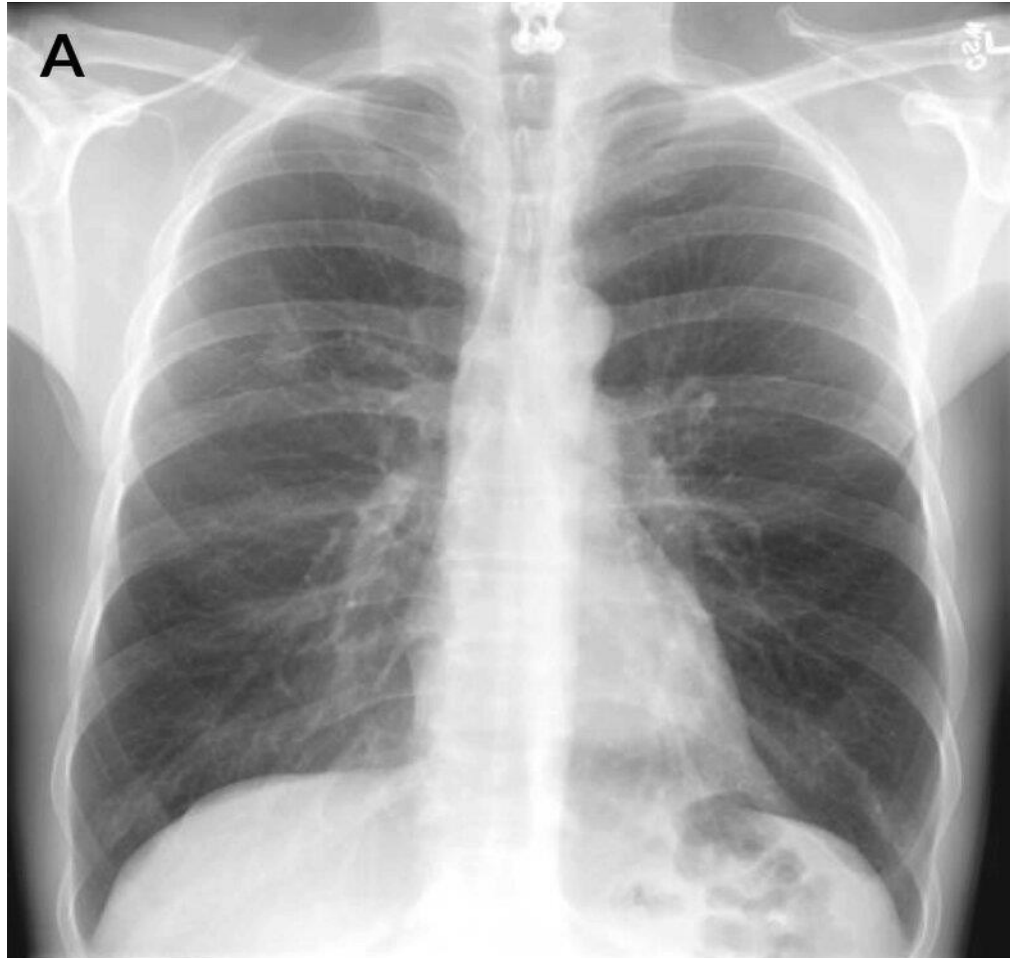
Lung Cancer

Diagnostics:

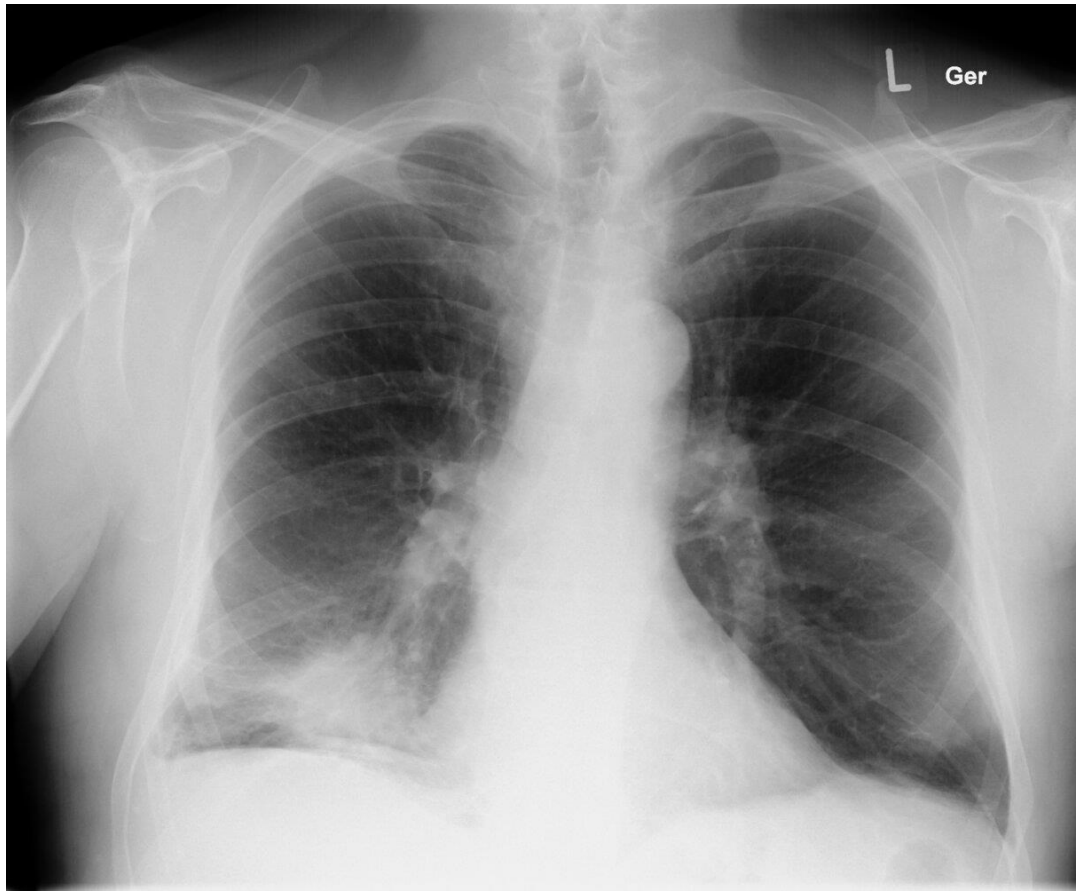
- CXR
- CT scan with IV contrast
- Fiberoptic bronchoscopy with endobronchial ultrasound
- Transthoracic needle biopsy
- Histopathological analysis
- Sputum cytology
- Genetic testing
- Immunohistochemistry
- Staging studies (MRI & PET CT)



Lung Cancer

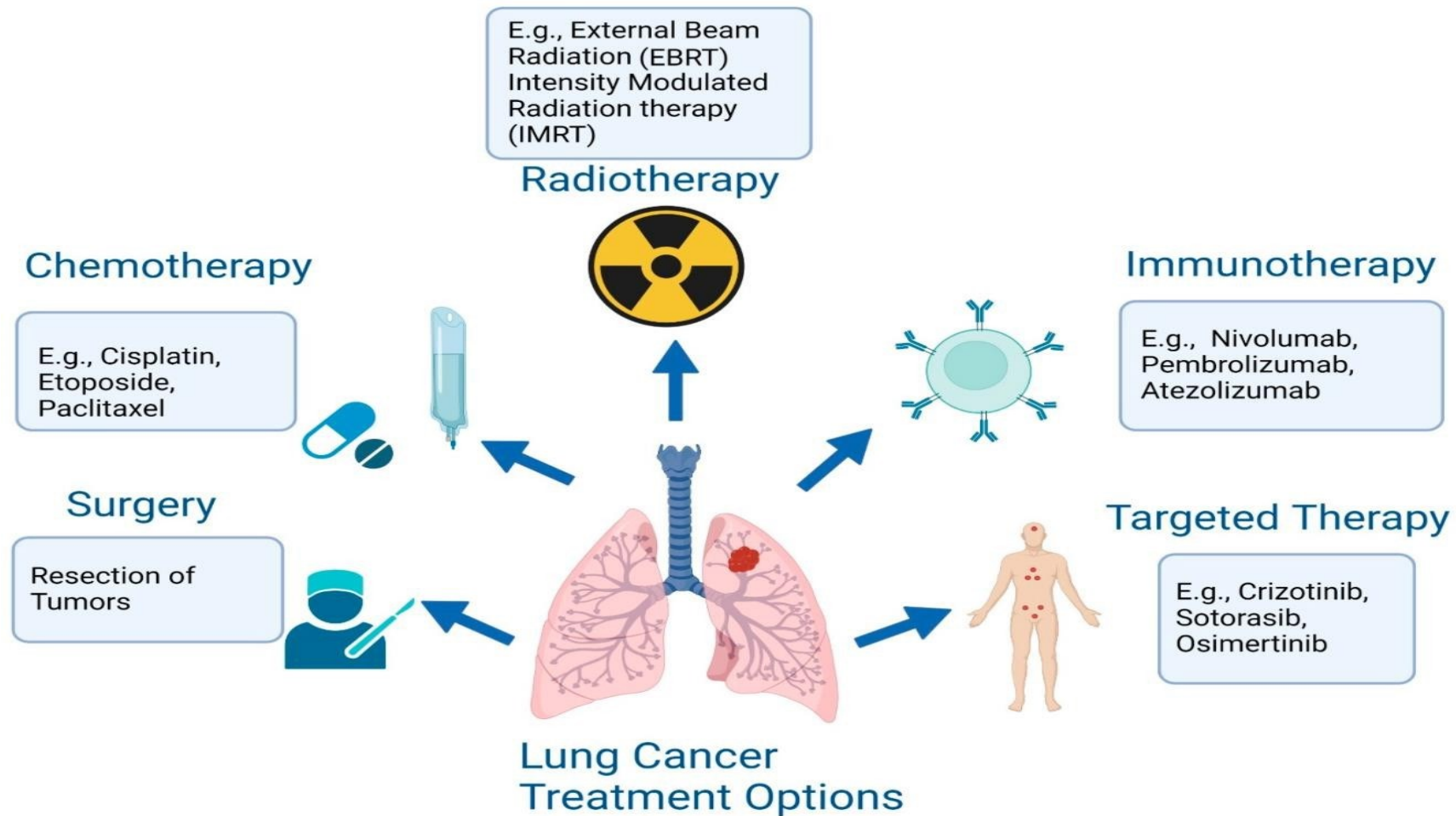


Lung Cancer



Lung Cancer

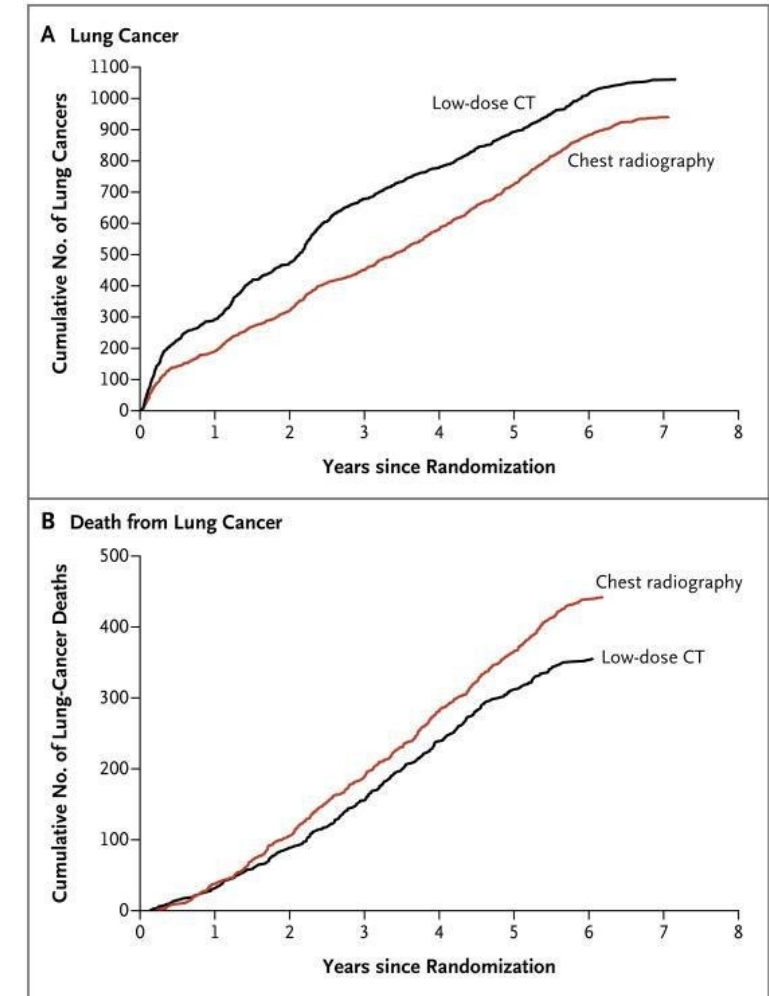
Management:



Lung Cancer

Screening:

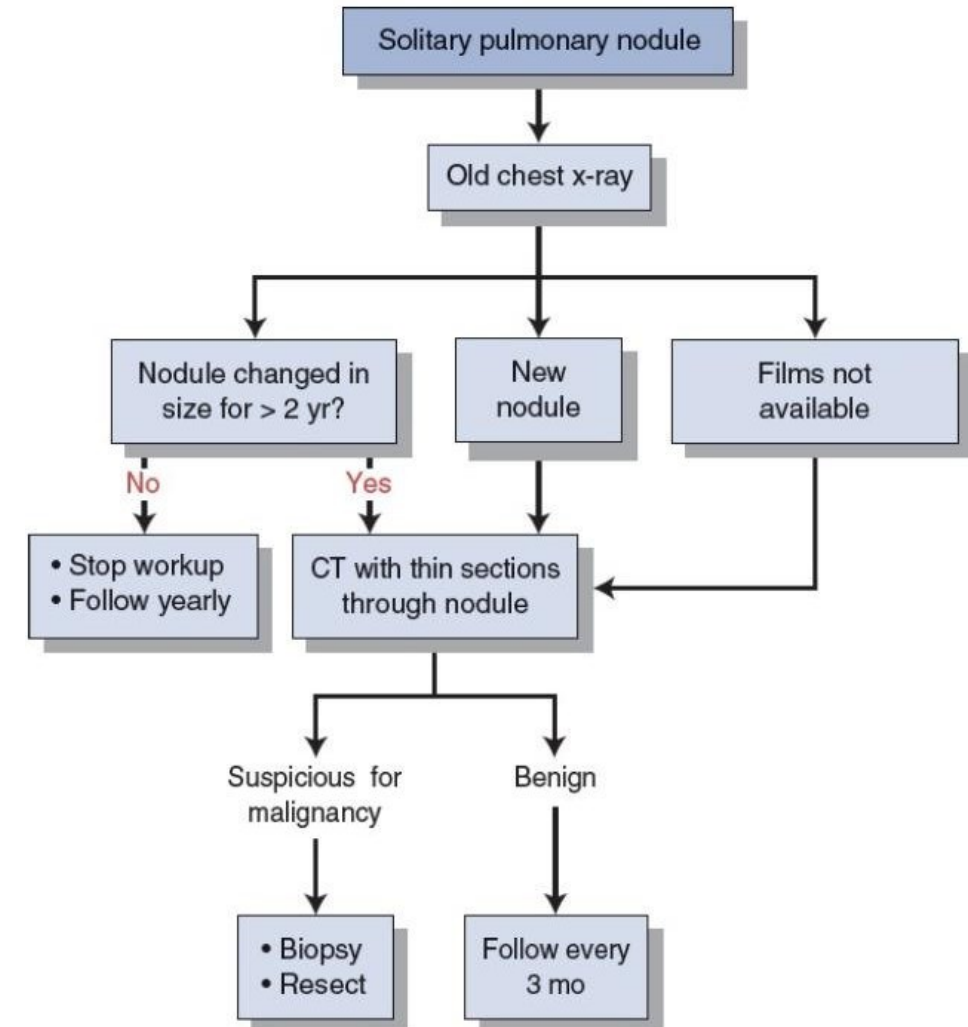
- Are ages 50 to 80 years.
- Have a 20 pack-year smoking history.
- Currently smoke cigarettes or quit within the past 15 years.
- Yearly low dose CT scan can reduce mortality by 20%



Solitary Lung Nodule

Approach:

- Review any previous imaging
- Obtain a thin-slice CT chest
- Identify nodule size and other characteristics:
 - Features suggestive of malignancy include irregular margins, larger size, upper lobe location
 - Features suggestive of other etiologies include calcifications



Approach to hemoptysis

Key History:

- Onset
- Dyspnea
- Cough/sputum production
- Fever
- Weight loss
- Symptoms of Heart disease
- Trauma
- Smoking history
- History of bleeding elsewhere
- Medications (anticoagulation)

Approach to hemoptysis

Key Physical Exam:

- Vital signs
- Pulse oximetry
- Exam of nasal mucosa
- Oropharynx
- Heart
- Lungs
- Lymph nodes and extremities (clubbing, cyanosis, edema)

Approach to hemoptysis

DDx:

RESPIRATORY

- Bronchial carcinoma
- Pneumonia
- TB (worldwide) (Fig. 31)
- Chronic bronchitis
- Bronchiectasis
- Pulmonary oedema
- Goodpasture's syndrome
- Wegener's granulomatosis

VASCULAR

- Pulmonary embolism
- Pulmonary hypertension – mitral stenosis
- Hereditary haemorrhagic telangiectasia

SYSTEMIC

- Coagulation disorders

General Workup:

- CBC
- CXR
- Coagulation profile (PT, PTT, INR)
- Sputum analysis
- ECG

Specific Workup:

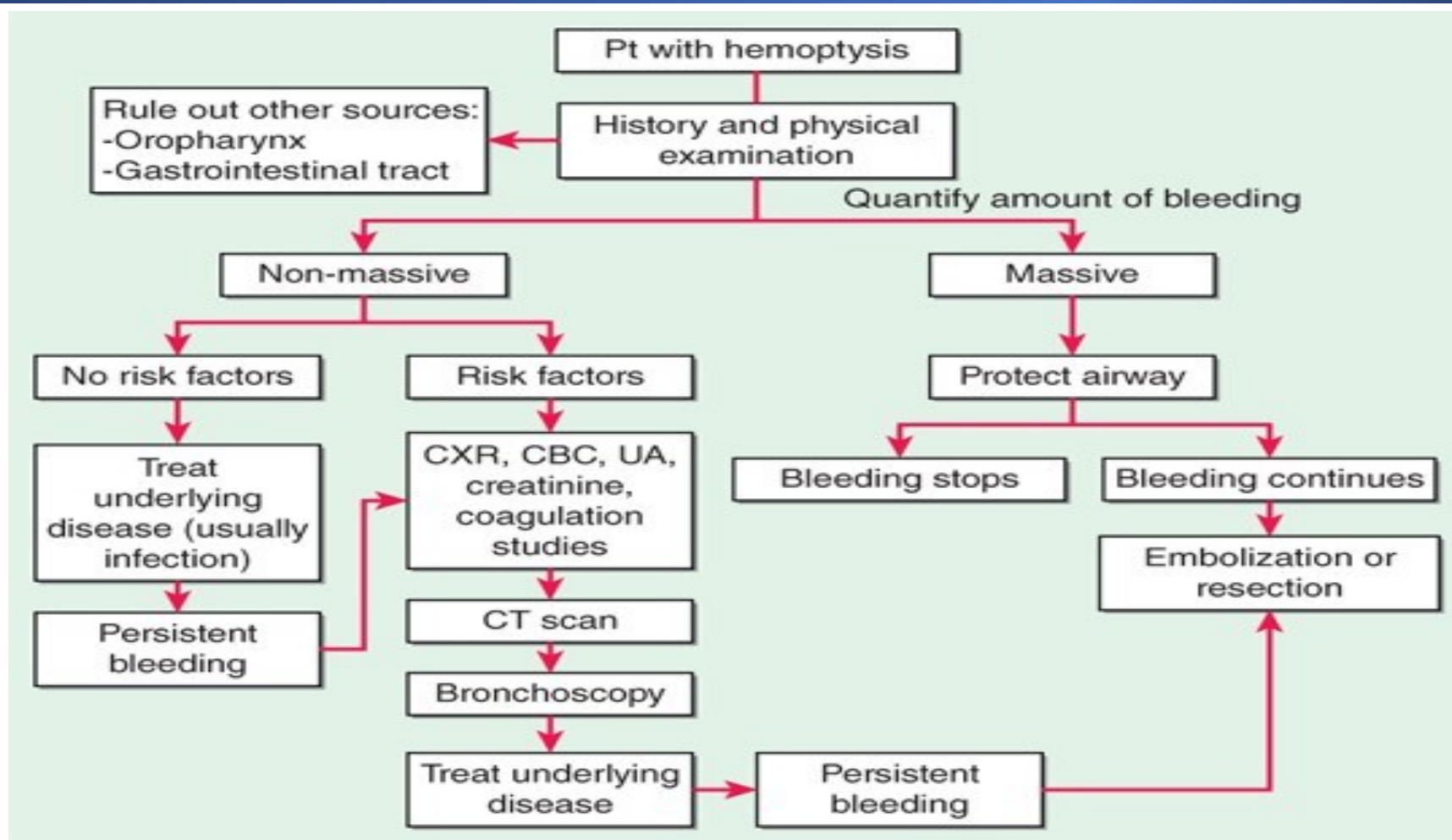
- CT scan
- TB testing
- Echo
- Bronchoscopy
- Basic metabolic panel (BMP)
- Urine analysis

Approach to hemoptysis

Management of Massive hemoptysis:

- Blood typing & cross-match
- IV Access
- Secure the airway
- Workup (CXR, coagulation studies, renal function testing, sputum testing)
- Control hemorrhage (TXA, FFP, platelets if <50 , RBCs as necessary, etc)
- Source control (depending on the underlying etiology)

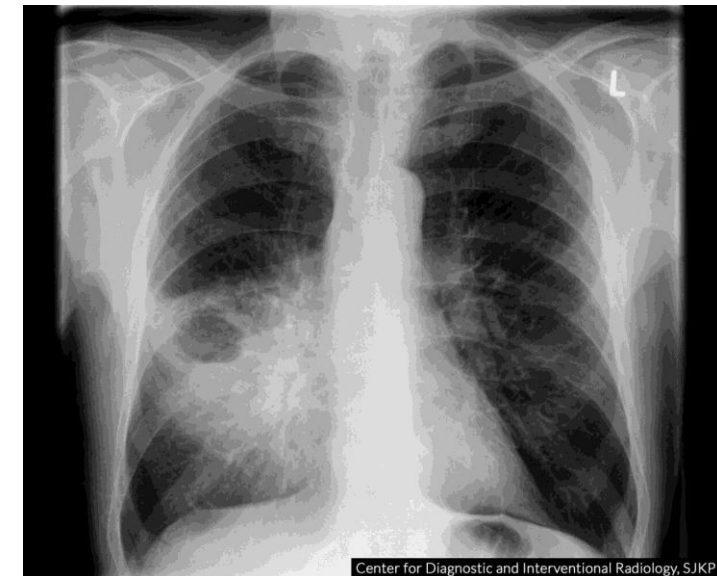
To Summarize



Case Scenario 1

A 69-year-old man comes to the physician because of a cough for the past 3 months. The cough is mostly dry, but the patient recalls a recent episode in which he coughed up mucus with dark red streaks. He has felt tired for the past few months. Over the past month, he has had poor appetite and a 5-kg (11-lb) weight loss. He also has occasional back pain, which has worsened in recent weeks. He has no recent travel history. He has smoked a pack of cigarettes daily for 30 years. He does not drink alcohol. His vitals are within normal limits. Auscultation of the lungs reveals wheezing in the lower right lung field. There is no peripheral lymphadenopathy. The liver is of normal size, and the spleen is not palpable. Laboratory studies show a hemoglobin concentration of 13.5 g/dL, serum calcium concentration of 12.3 mg/dL, and a total serum protein of 7.0 g/dL. X-rays of the chest are shown. Which of the following is the most likely diagnosis?

- a. Multiple myeloma
- b. Tuberculosis
- c. Lobar pneumonia
- d. Small cell lung carcinoma
- e. Sarcoidosis
- f. Squamous cell lung carcinoma



Case Scenario 2

A 50-year-old man comes to the physician after a fall on a flight of stairs earlier that day. He slipped, causing him to fall and hit the handrail with the lateral part of his chest. He has no shortness of breath. He returned from a 10-day trip to South Africa 4 weeks ago. He smoked one pack of cigarettes daily for 10 years but stopped 18 years ago. He drinks one to two beers a week. His temperature is 37.1°C (98.8°F), pulse is 78/min, and blood pressure is 126/72 mm Hg. Examination shows normal breath sounds bilaterally. The left lateral chest wall is tender to palpation. An x-ray of the chest shows no fracture; there is a 5-mm lesion in the right upper lung field. There are no previous x-rays of the patient available. Which of the following is the most appropriate next step in management?

- a. Follow-up x-ray of the chest in 12 months
- b. Follow-up x-ray of the chest in 6 months
- c. Interferon- γ release assay
- d. Positron emission tomography
- e. CT scan of the chest
- f. Sputum culture
- g. Reassurance
- h. Bronchoscopy with biopsy

Questions?



References & further reading

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine
20th edition
- Amboss
- Cecil Essentials of Medicine 10th edition
- First Aid for the internal medicine board
4th edition
- Osmosis

Thank You